WAZP

MONOPHONIC SYNTHESIZER

Owner's Manual

Version 1.0.8



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Welcome

WAZP is a monophonic synthesizer plug-in inspired by the small, buzzy British synths of the late 1970s: two raw oscillators, a screaming state-variable filter, a control oscillator for wobbles and random bleeps, and two envelopes that can repeat on their own.



Formats

- macOS: Audio Unit, VST3 and standalone app (Apple Silicon and Intel), one .pkg installer.
- Windows: VST3 and standalone app, one Setup.exe installer.
- Linux: VST3, LV2 and standalone app (.tar.gz with an install script).

System Requirements

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macOS	macOS 10.13 High Sierra or newer on Intel; macOS 11 Big Sur or newer on Apple Silicon (M1 and later, runs natively). A host that loads Audio Unit (AU) or VST3 instruments, e.g. Ableton Live, Logic Pro, GarageBand, Bitwig Studio, Reaper, FL Studio, Cubase, or the included standalone app.
Windows	Windows 10 or 11, 64-bit. A host that loads VST3 instruments, or the included standalone app.
Linux	64-bit (x86_64) distribution with ALSA or JACK. A host that loads VST3 or LV2 instruments, or the included standalone app.
All systems	64-bit host, about 30 MB of disk space, screen of at least 1280 × 800 (the panel can be resized), MIDI keyboard or MIDI clips to play it.

Installing

macOS: open *WAZP-1.0.8-macOS.pkg* and follow the steps. Click **Customize** to choose formats.

⚠ **macOS security warning.** The WAZP installer is not notarised by Apple, so the first time you open it macOS may say it “cannot be opened because it is from an unidentified developer” or “cannot be verified”. This is expected. To install: **right-click** (or Control-click) the .pkg and choose **Open**, then click **Open** again. If macOS still blocks it, go to **System Settings → Privacy & Security**, scroll down and click **Open Anyway** next to the WAZP message.

Windows: run *WAZP-1.0.8-Windows-Setup.exe*. The VST3 goes to *C:\Program Files\Common Files\VST3*.

Linux: unpack the .tar.gz and run *./install.sh*.

Then rescan plug-ins in your DAW (Ableton Live: Settings → Plug-Ins → Rescan) and load WAZP on a MIDI track.

Ableton Live: if Live once failed to open a WAZP version, it may keep it disabled (“Some plug-ins are disabled”). After installing the new version, hold **Alt/Option** and click **Rescan** in Settings → Plug-Ins to rescan everything, then load WAZP again.

WAZP is monophonic: one note at a time, last-note priority. Hold a note and play another for a legato line; with GLIDE up the pitch slides between them.

The Panel

Drag a knob up or down to turn it, double-click to return it to its default. There are no pop-up hints when you hover over the panel: while you move a control, the preset window shows its name and value. The window can be resized from the bottom-right corner.

Keyboard

Control	What it does
SYNC	Hard-syncs oscillator 2 to oscillator 1. Set oscillator 2 higher than oscillator 1 (range or PITCH) and sweep PITCH for the classic tearing sync sound.
HOLD	Keeps the last note sounding after you let go of the key. Playing a new note replaces it; switching HOLD off releases it. Great with the REPEAT envelopes for self-playing patterns.
GLIDE	Portamento: how long the pitch takes to slide to a new note. Fully left = off, up to about 2 seconds.
Keys	The on-panel keyboard is playable with the mouse (C3 - C5) and lights up with incoming MIDI notes. MIDI pitch bend (+/- 2 semitones) and the mod wheel (adds pitch modulation) also work.

Oscillators

Control	What it does
FT. (both)	Footage / octave range: 32′, 16′, 8′, 4′, 2′. 8′ plays at the pitch of the key.
WIDTH %	Pulse width of oscillator 1, from 10 % to 90 % (50 % is a square wave). Only heard with the pulse waveform.
Wave (both)	OFF, pulse (⊓) or sawtooth. OFF silences the oscillator.
PITCH	Tunes oscillator 2 against oscillator 1, +/- 12 semitones. Small offsets give a fat detuned sound; 7 = a fifth, 12 = an octave.

Mixer / Output

Control	What it does
VOLUME	Master output level, -80 dB to +6 dB.
OSC 1, OSC 2	Level of each oscillator going into the filter (0 - 10). Push both up for more filter drive and grit.

The Panel (continued)

Control Osc.

Control	What it does
FREQ.	Speed of the control oscillator, 0.05 Hz up to 1 kHz. It restarts with every new note. Above about 30 Hz it stops being a wobble and becomes audio-rate modulation: metallic, growling, FM-like tones.
PITCH MOD.	How much the control oscillator moves the pitch of both oscillators (up to +/- 12 semitones). Vibrato at low settings, sirens and bleeps higher up.
FREE / SYNC	Slide switch under FREQ. FREE: FREQ. sets the speed in Hz and the control oscillator restarts with every note. SYNC: the control oscillator locks to your DAW tempo and FREQ. steps through note lengths, from 4 BARS (fully left) through 1 BAR, 1/2, 1/4, 1/8 and 1/16 (with dotted and triplet values) to 1/32 (fully right); the display shows the chosen value. While the DAW is playing, the cycle also follows the song position, so wobbles and random steps land on the beat.
OSC 1 / BOTH / OSC 2	Slide switch under PITCH MOD.: chooses which oscillator the pitch modulation (control oscillator and mod wheel) moves. BOTH moves both together. OSC 1 or OSC 2 moves only one, so it beats and drifts against the steady one: moving detune, chorus-like beating and, with SYNC on and OSC 2 selected, sweeping sync tones.
NOISE SIGNAL	Level of white noise in the mixer (0 - 10). Use it alone with the filter for wind, surf and percussion.
Wave	Shape of the control oscillator: sine, square, noise (smoothed random movement) and RND (a new random value every cycle: sample and hold).

Filter

Control	What it does
FREQ.	Filter cutoff frequency, 20 Hz to 20 kHz.
Q	Resonance. The filter is two stages in series, so it rings hard and screams. At the top it sits right at the edge of self-oscillation: a piercing, singing resonance on every note, at any cutoff.
CONTROL	Filter type: LO (low-pass), BAND (band-pass), HI (high-pass) or NOTCH (cuts a band around the cutoff and lets everything else through; hollow, phasey tones, and a phaser-like effect when swept by the control oscillator).
Left -/+ knob	Amount of the control oscillator on the cutoff (wah, wobble, random filter steps), up to +/- 8 kHz. With a low FREQ. setting a deep amount can close the filter completely on the low part of the cycle, giving a gated, chopped sound. Centre = no modulation.
Right -/+ knob	Amount of the CONTROL ENV. on the cutoff. Right of centre opens the filter, left of centre closes it (inverted envelope).

Envelope Generators

Control	What it does
VCA ENV. ATTACK / DECAY	How fast the sound rises when you press a key and how fast it

Presets

The preset window at the top right shows the current preset number and name.

Control	What it does
< >	Step through the factory presets, followed by your own presets.
Preset display	Click the name to open the full preset list: all factory presets, then your own presets (MY PRESETS). The current one is ticked; click any name to load it. OPEN PRESET FOLDER at the bottom shows your presets in Finder / Explorer.
SAVE	Saves the current sound as a .wazp file. By default it goes to *Documents / Transistor Labs / WAZP / Presets*; presets saved there appear in the < > list.
LOAD	Opens any .wazp file.

WAZP comes with 33 factory presets: INIT, BUZZ BASS, WASP STING, SYNC SCREAM, REPEAT PERC, DREXY ACID, NOISE WIND, SIREN, RANDOM BLEEPS, HOLLOW LEAD, HIGH PASS FIZZ, BAND PLUCK, SUB DRONE, WOBBLE, SLOW SWELL, BEATING DETUNE, SYNC SWEEP, NOTCH PHASER, FM BELL, OSC 2 FM GRIT, CLANG, VIBRATO OSC 1, RING TONE, AUDIO RATE FILTER, KICK, TOM, CLOSED HAT, OPEN HAT, METAL PERC, RIM SHOT, SNARE TIGHT, SNARE NOISY and 8-BIT SQUARE.

The drum presets (KICK to SNARE NOISY) are short one-shots: play them from MIDI clips like a drum machine, and change the pitch by playing higher or lower notes. The presets also appear in your DAW's program list.

> Your DAW saves the complete state (every knob and the preset name) with the project, so you don't need to save a preset to keep a sound in a song.

Automation & MIDI

All controls can be automated and MIDI-mapped from your DAW. WAZP responds to MIDI notes on any channel, pitch bend (+/- 2 semitones) and the mod wheel (CC 1, adds pitch modulation from the control oscillator). Velocity is ignored, as on the original instrument.

Sound Design Tips

- **Classic buzz:** both oscillators on sawtooth, PITCH slightly off zero, Q around 50 %, a short CONTROL ENV. on the filter.
- **Sync lead:** SYNC on, oscillator 2 one range higher, then automate PITCH.
- **Self-playing rhythm:** VCA REPEAT fully left with a short DECAY gives repeating notes; add CONTROL ENV. REPEAT for moving filter pulses.
- **Bleeps:** control-osc wave on RND, FREQ. around 6 - 10 Hz, PITCH MOD. up.
- **Noise percussion:** oscillators OFF, NOISE SIGNAL up, BAND filter with high Q, short VCA decay.

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